VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Application ID: [Enter ID]

VergeLink — camera-free eye + jaw-clench control of fan and light for people with quadriplegia

Team ID: [Team ID] | Track: [Track] | Dept. of Biomedical Engineering, VCET Erode (Batch 23BM7A)

Faculty guide: Mrs. C. Radhika · Industry mentor: K. P. Amalnath, Mediowel Healthineers

Problem Context (1/2): Who Is Affected

~20,000 / yr
NEW SPINAL CORD INJURIES, INDIA

1.5 million
INDIANS LIVING WITH SCI

54.4 lakh
MOVEMENT DISABILITY (CENSUS 2011)

~12.9 h / day
UNPAID CARE, SCI USERS (US STUDY)

Project Overview

- **Title:** VergeLink — camera-free eye + jaw-clench control of fan and light.
- **Team / Track:** [Team ID] | [Track]
- **Core innovation:** bilateral EOG vergence + ToF depth + a deliberate jaw clench (EMG) give two-factor, camera-free appliance control from ₹13,093 of parts.

Problem Statement

People with quadriplegia who can still move their eyes and jaw cannot switch their own fan or light. Every change waits on a caregiver, and the hands-free options available depend on a camera, on clear speech, or cost well beyond most Indian households.

Target Beneficiary Segment

- **Primary users:** adults with high cervical spinal cord injury, ALS/MND or similar conditions who keep reliable eye movement and jaw control.
- **Secondary users:** family caregivers at home, and attendants on rehabilitation wards who handle every environmental request by hand.
- **Setting:** Indian homes and wards with ordinary ceiling fans, regulators and wall switches — no smart-home wiring assumed.
- **Who is out of scope (for now):** users without dependable jaw control, and wheelchair driving — that needs separate safety engineering.

Data sources: International Health Policies (SCI incidence and prevalence), Census of India 2011 (movement disability), SCIRE Project / Robinson-Whelan & Rintala 2003 (care hours, US data) — see References.

Problem Context (2/2): Why Today's Options Fall Short

Existing hands-free approaches vs the needs of a person with quadriplegia

Approach	What it needs from the user	Where it breaks down at home	Cost signal
Calling a caregiver	Voice, and a caregiver within earshot	Every change waits on someone else, day and night	Caregiver time
Camera eye-tracker	Stable head, screen-facing camera, steady lighting	Lighting changes, privacy concerns, needs a screen in view	Imported devices
Voice assistant	Clear, consistent speech	Fails with weak, slurred or absent speech	Low
Sip-and-puff / switch	Breath control or residual movement	Mouth-held tube, caregiver set-up, hygiene	Moderate
EOG communicator (Netravaad)	Eye gestures	Built for communication, not for direct appliance control	₹1.65 lakh
VergeLink (this project)	**Eye movement + jaw clench**	**Prototype stage — performance to be measured in Stage 3**	**₹13,093 parts**

The gap

No low-cost, camera-free option lets the user switch a specific appliance with intent they control — without speech and without a caregiver.

Public search disclosure

VergeIO (2025) showed bilateral EOG can tell 4 depth gestures apart on glasses (97%). It is an input study, not home-appliance control for paralysed users.

Positioning: VergeLink combines published bilateral-EOG vergence sensing with ToF depth evidence and a jaw-clench confirmation, aimed at one job: independent fan and light control.

User Journey (1/2): Before State

BEFORE: every fan or light change needs a caregiver



Team diagram, qualitative before-state flow.

Figure: Current pain-point flow for one fan or light change.

Baseline we will record before Stage 3

- 3-day log with a partner family: each fan/light request, time asked, time done, who helped.
- Same log format reused in Stage 3 trials, so before and after are measured the same way.

Quantified Inefficiency

100%
OF FAN / LIGHT CHANGES NEED ANOTHER PERSON

~12.9 h / day
UNPAID CARE RECEIVED BY SCI USERS (US STUDY)

₹1.65 lakh
PRICE OF AN INDIAN EOG COMMUNICATOR

- **Time loss:** user waits for every change; caregiver's own task is interrupted each time.
- **Safety:** a user left too hot or in the dark cannot fix it alone.
- **Wait time per change:** not yet measured — the baseline log above will give it.

Honesty note: the 100% figure follows from the user being unable to reach a switch; the care-hours figure is US data; local wait times will come from our own baseline log.

User Journey (2/2): After State & Efficiency

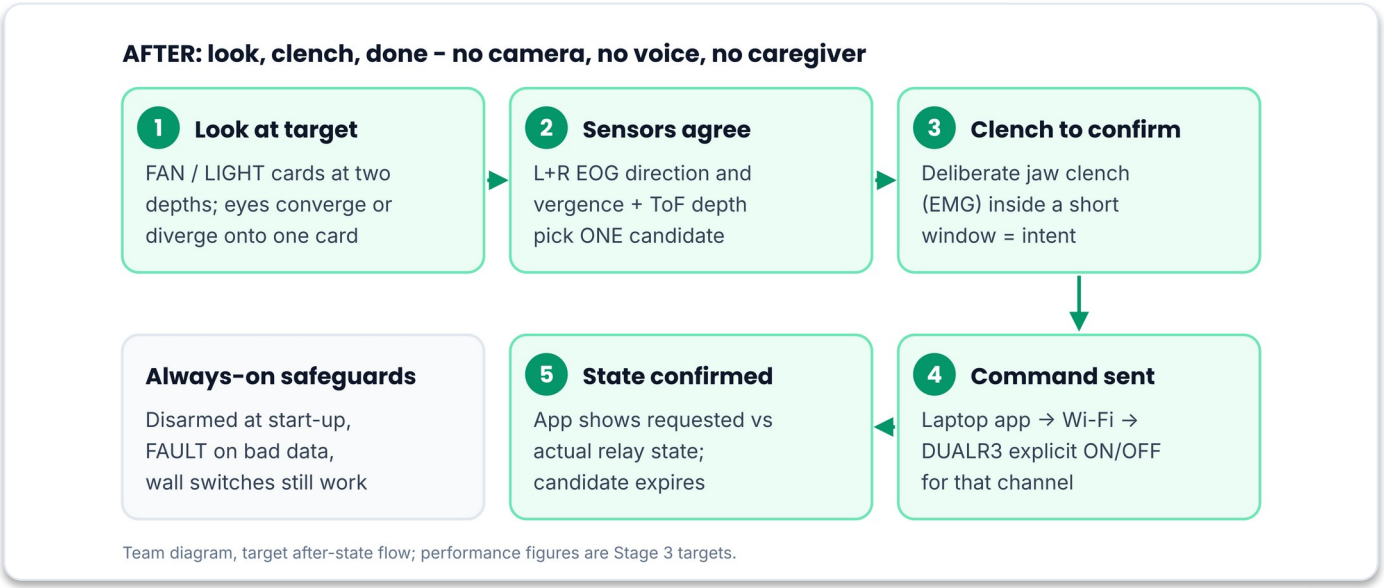


Figure: Target after-state flow — look, clench, relay switches.

How these targets will be measured (Stage 3)

- **Timed trials:** app logs time from target shown to relay state confirmed, per attempt.
- **False activations:** 30-minute armed sessions of rest, talking, chewing and blinking.
- **Accuracy:** 2 targets × 2 depths, scripted order, team volunteers under approval.

Quantified Efficiency (targets)

Metric	Before	VergeLink target
Who switches	Caregiver	User alone
Time per change	Caregiver response	≤ 3 s look → relay
Hands-free cost	₹1.65 L (Netravaad)	₹13,093 BOM (−92%)
Camera / mic	Needed	Neither
False activations	n/a	≤ 1 per 30 min
Selection accuracy	n/a	≥ 90% (2 × 2 targets)
Caregiver trips	Every change	Zero for fan/light

Scope: only the cost row is a fact today. Every other VergeLink figure is a design target to be measured after parts arrive.

Beneficiary impact: routine fan and light changes become self-directed, so the caregiver's time goes to care that genuinely needs a person.

System Architecture (1/2): Block Diagram

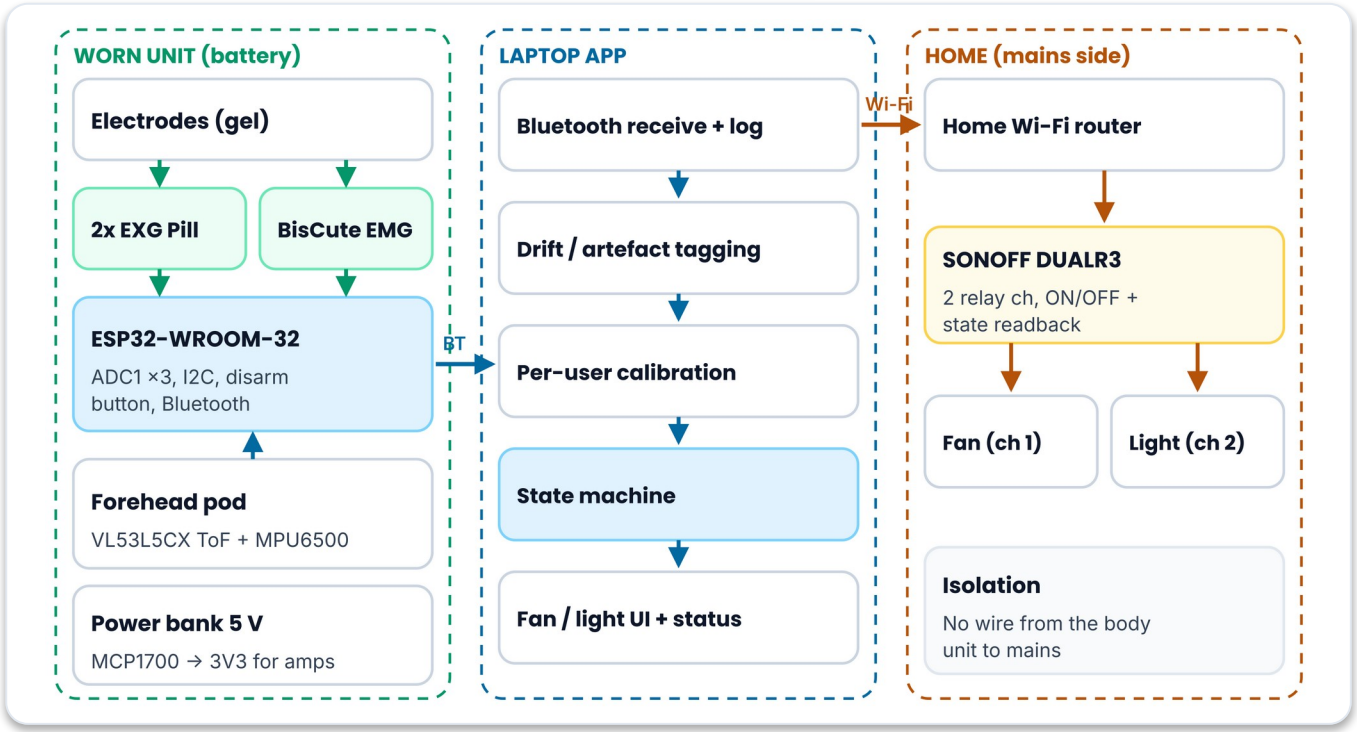


Figure: Subsystems — worn sensing unit, laptop application, and mains-side relay.

Signal path: electrodes → bio-amps → RC filter → ESP32 ADC1 → Bluetooth → laptop decoder → Wi-Fi → DUALR3 relay → fan / light.

Subsystems & Firmware / Logic Flow

- **Mechanical:** PETG body-worn enclosure + forehead sensor pod on an elastic headband.
- **Power:** 10,000 mAh bank → ESP32 VIN; separate MCP1700 3V3 rail for the bio-amps.
- **Electronics:** 2× EXG Pill, BisCute, ESP32 ADC1, VL53L5CX + MPU6500 on I2C.
- **Firmware:** timed sampling of 3 analog channels + I2C sensors; every frame time-stamped and health-flagged.
- **Laptop logic:** DISARMED → ARMED → CANDIDATE (gaze + depth agree) → CONFIRMED (clench in window) → explicit ON/OFF.

Architecture rationale: analog inputs use ADC1 only because ESP32 ADC2 cannot be used while the radio runs; the body-worn unit has no wired path to mains.

System Architecture (2/2): Specs & Hardware Selection

Engineering Target Specifications

Parameter	Target spec	Status
Eye signals	2 ch bilateral horizontal EOG	Parts ordered
Intent confirm	1 ch jaw EMG (72–720 Hz band)	Parts ordered
Depth evidence	8×8 zones, up to 4 m	Parts ordered
Sampling	≥ 500 Hz per analog channel	Target
Laptop link latency	< 50 ms per frame	Target
Look → relay time	≤ 3 s	Target
Session length	≥ 8 h on battery	Target
Appliance channels	2, explicit ON / OFF	Parts ordered

Hardware Selection & Protocols

Block	Selected	Why
EOG front-end	2× BioAmp EXG Pill	Single-supply, matches published bilateral set-up
EMG front-end	BisCute	Built-in EMG band-pass; low cost
Controller	ESP32-WROOM-32	3 ADC1 inputs + Bluetooth + Wi-Fi
Depth	VL53L5CX (I2C)	8×8 zones, Class 1 emitter
Head motion	MPU6500 (I2C)	Motion rejection only
Laptop link	Bluetooth serial	No extra radio hardware
Relay	SONOFF DUALR3	2 channels in one retrofit box

Superseded choices: ADS1293 AFE (₹6,995, cancelled), nRF52840 MCU and two MINIR4 relays from Stage 1 were replaced to stay within the ₹16,000 cap.

CAD Design & Mechanical Specs (1/2)

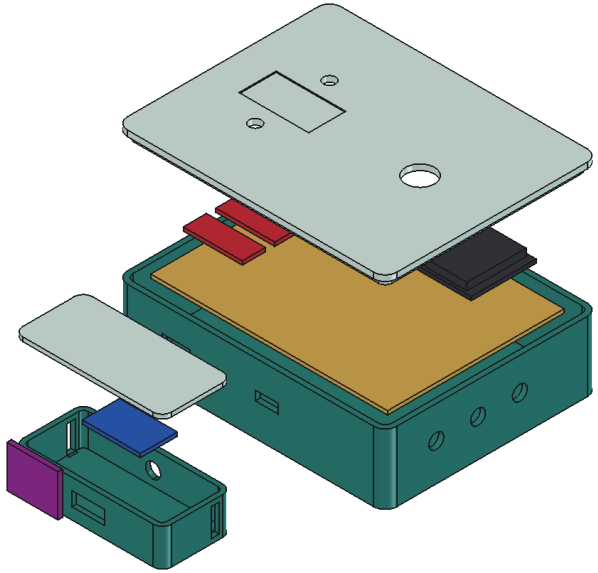
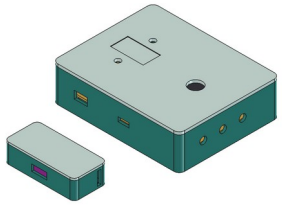


Figure: FreeCAD 1.1 exploded assembly of enclosure, lid, perfboard, modules and sensor pod.



12 parts
Parametric FreeCAD model; STEP exported for printing.

Mechanical Specifications

Item	Specification
Enclosure (outer)	116 × 92 × 30 mm, 2.2 mm walls, 5 mm radii
Board mounting	4 × M2.5 standoffs, 3×4 in perfboard pattern
Wall ports	3 electrode leads, pod cable, USB power, ESP32 USB
Lid	Disarm button hole, 2 status LEDs, label pocket
Forehead pod	64 × 30 × 16 mm; 14 × 10 mm ToF window
Headband slots	26 × 3.5 mm, elastic band
Strap loops	2 rear loops for 25 mm webbing
Material	PETG, FDM (college printer)

Design iteration: CAD review caught the strap-loop gap modelled on the wrong side (loop was solid against the case) — corrected before FEA.

CAD Design & Mechanical Specs (2/2): FEA

Peak 21.3 MPa (solver max) · max displacement 0.20 mm · 50 N pull · deformation ×10

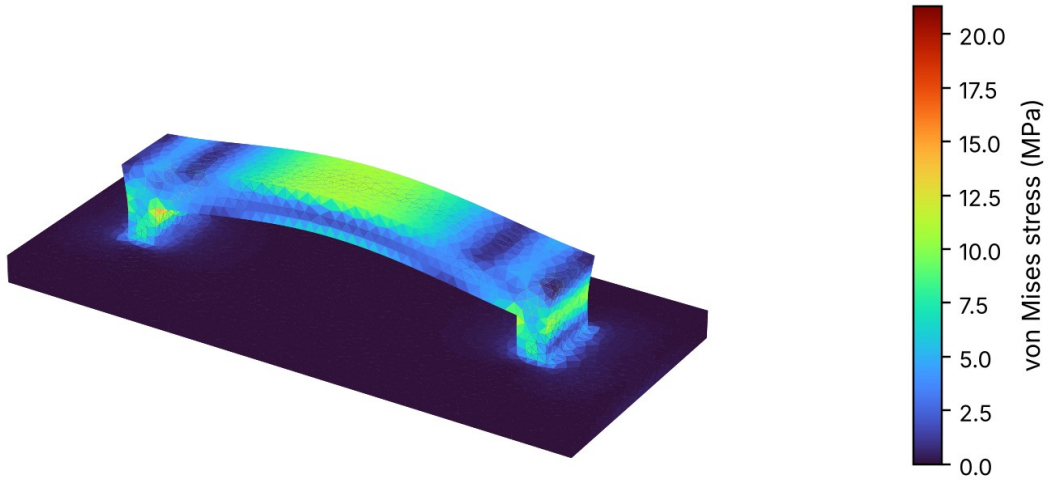


Figure: CalculiX von Mises stress on the strap loop, 50 N pull, deformation ×10.

Why this part: the strap loop carries the whole box if the strap snags on a wheelchair or bed rail.

FEA Load Case & Results

- **Geometry:** 30×6×6 mm loop, 2 mm posts, 3 mm bridge, on a 2.2 mm wall patch.
- **Load / support:** wall face fixed; 50 N pull on the bridge, away from the case.
- **Mesh:** 44,420-node 2nd-order tetrahedra; E = 2.1 GPa, v = 0.38.
- **Result:** peak 21.3 MPa at the sharp bridge/post corner; max displacement 0.20 mm; FoS ≈ 2.1–2.3 vs ~45–50 MPa PETG yield.

Material Selection

Material	For this device	Decision
PETG	Tough, prints well, tolerates warm rooms	Selected
PLA	Softens at modest temperatures, brittle	Rejected
ABS	Warps on an open-frame printer	Rejected

Next action: add a 1 mm fillet at the stress corner and print with layers along the load path, since FDM inter-layer strength is below bulk.

Electronics Schematics & BOM (1/2): Schematic

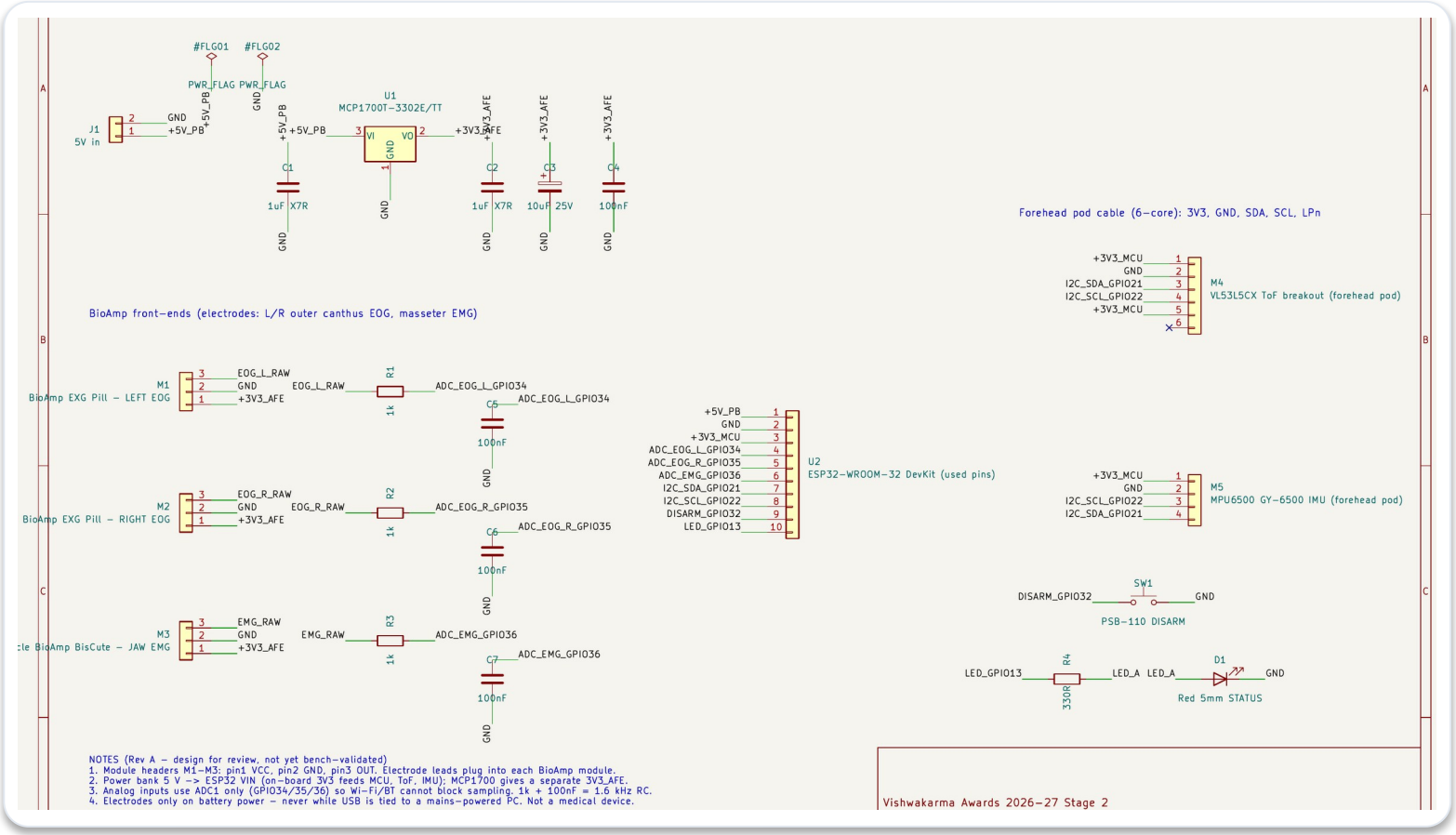


Figure: KiCad 10 schematic, Rev A — ERC 0 errors / 0 warnings (kicad/VergeLink_Mainboard.kicad_sch).

ESP32 Pin Map

Pin	Signal
GPIO34 (ADC1_CH6)	Left EOG
GPIO35 (ADC1_CH7)	Right EOG
GPIO36 (ADC1_CH0)	Jaw EMG
GPIO21	I2C SDA
GPIO22	I2C SCL
GPIO32	Disarm button
GPIO13	Status LED (330 Ω)
VIN	5 V from power bank
3V3 out	ToF + IMU supply

Power distribution: power bank 5 V → ESP32 VIN; MCP1700 LDO (1 µF in/out, 10 µF + 100 nF bulk) feeds only the three bio-amps; 1 kΩ + 100 nF RC (1.6 kHz) on each ADC input.

Electronics Schematics & BOM (2/2): Itemized BOM

Itemized Prototype Bill of Materials (as ordered, incl. GST & shipping)

Component	Part / specification	Qty	Vendor (India)	Order status (6 Oct 2026)	Cost ₹
EOG front-end	Upside Down Labs BioAmp EXG Pill kit	2	Upside Down Labs	Paid, order received	5,198.00
EMG front-end	BisCute kit + 100 cm cable + 2 electrode packs	1	Upside Down Labs	Paid, order received	1,346.00
Wi-Fi relay	SONOFF DUALR3, 2 channel	1	SONOFF India	Paid	1,665.06
Power bank	Portronics Aero 10, 10,000 mAh	1	Portronics	COD, confirmed	1,303.00
Depth sensor	SmartElex VL53L5CX ToF breakout	1	RoboTechMinds	Paid	1,096.80
MCU	ESP32-WROOM-32 38-pin DevKit	1	Robu	Completed	421.00
IMU	GY-6500 MPU6500	1	FlyRobo	COD, confirmed	345.00
LDO	MCP1700T-3302E/TT, SOT-23	2	Robu	Paid, processing	145.00
Passives & build	Resistor kit, caps, LEDs, perfboard, headers, wire, button, OLED, cables	lot	Robu / IHC / ElectronComponents	Paid / completed	1,573.41
Total				**Cap ₹16,000 → headroom ₹2,906.73**	**13,093.27**

Procurement: every part is from Indian sellers — no imports. Delivery inventory and module-revision checks are in progress, so no lead time is claimed as met yet.

Your Maker Journey (1/2): Stage 1 → Stage 2

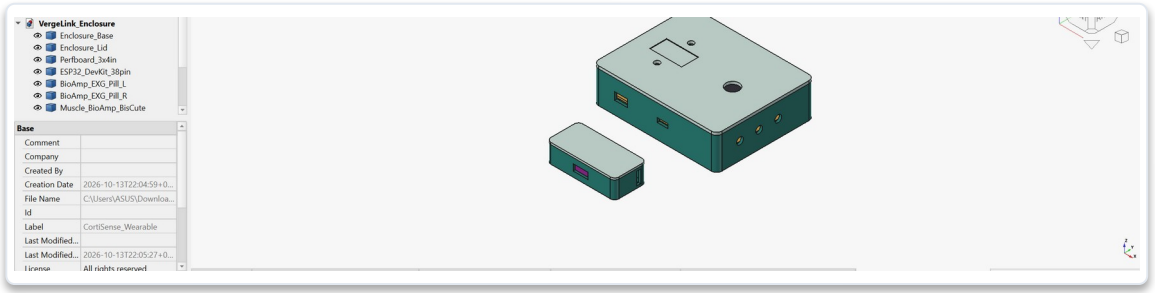


Figure: FreeCAD 1.1 session — VergeLink_Enclosure feature tree and model.

[Add photo]
Team members soldering the perfboard / assembling the enclosure, wearing institutional ID badges.

Stage 1 to Stage 2 Engineering Iterations

- **Front-end pivot:** Stage 1 planned ADS1293 + AD8232 + nRF52840. The ₹6,995 ADS1293 order was cancelled; two EXG Pills + BisCute + ESP32 keep the build inside ₹16,000.
- **Scope made honest:** vertical EOG channel, magnetometer heading, wheelchair DAC and dry electrodes are out of this build; Stage 1 simulation figures are not reported as results.
- **Actuation:** two MINIR4 relays replaced by one SONOFF DUALR3.
- **CAD milestone:** 12-part FreeCAD assembly, STEP export and CalculiX FEA completed 14 Oct 2026.
- **EDA milestone:** KiCad 10 schematic, 21 components, 15 named nets; ERC 0 errors / 0 warnings on 14 Oct 2026.

Institutional sign-off: Faculty guide Mrs. C. Radhika, Dept. of Biomedical Engineering, VCET Erode (Batch 23BM7A) — [add review date / signature].

Your Maker Journey (2/2): Evidence & Repository

VERGELINK · STAGE 2

Raw Repository Linkage

github.com/amalnathkp13-boop/vergelink

```
presentation/   deck (.pptx) + PDF
cad/            .FCStd, .step, FEA model + results
kicad/          .kicad_sch, netlist, BOM, ERC report
figures/        every figure used in this deck
README.md       status, FEA summary, safety notes
```

[Add photo]
Parts laid out on delivery, with order labels visible.

[Add photo]
Joint review with the faculty guide.

Work-in-Progress Software Evidence

- **CAD:** parametric FreeCAD tree with unsuppressed features (slide 12 screenshot).
- **EDA:** KiCad ERC report — 0 errors, 0 warnings (kicad/erc_report.rpt).
- **FEA:** 44,420-node mesh and CalculiX result stored in the .FCStd file.

Authenticity Verification Plan

- Team members commit their own build logs, photos and firmware to the repository as each milestone lands.
- All photos taken with institutional ID badges in frame; no embedded videos — links only.

Timestamps: native files carry their creation dates; each new commit adds a dated record of Stage 2 progress for audit.

Safety & Hazard Mitigation

FMEA Matrix — failure modes and embedded protections

Failure mode	Potential hazard	Embedded fail-safe / mitigation response
Electrodes worn while USB goes to a mains PC	Leakage current through skin electrodes	Battery-only operation whenever electrodes are attached; program over USB only with electrodes removed; no wired path to the relay.
Chewing / talking produces jaw EMG	Fan or light switches unintentionally	EMG accepted only inside a candidate window after gaze + depth agree; refractory period; physical disarm button (GPIO32).
EOG drift, electrode peel, sensor dropout	Wrong target selected	Clipping / stale / missing-sensor flags force FAULT; per-session calibration; candidates expire automatically.
Bluetooth or Wi-Fi link drops	Command lost or repeated	Restart in DISARMED; explicit ON / OFF (never toggle); app shows requested vs actual relay state.
Supply sag / LDO overload	Noisy or clipped bio-signals	Bio-amps on a separate MCP1700 rail; ESP32 and radio on the DevKit regulator; rail currents measured at bring-up.
Mains wiring of the DUALR3	Shock or fire	Licensed electrician installs in existing switchboard; wall switch and fan regulator stay in service.

Physical Fail-safes

- **Disarm button:** momentary PSB-110 on GPIO32 drops the system to DISARMED at once.
- **Enclosures:** PETG case and pod cover all boards; no exposed conductors near the skin.
- **Isolation:** body-worn unit and mains relay share no wire — only Wi-Fi.

Standards Compliance (considered, not certified)

- **ISO 14971:** risk-management file started from this FMEA.
- **IEC 60601-1:** leakage-current principles for body-worn electrodes.
- **IEC 60825-1:** VL53L5CX emitter is Class 1 (per ST).

Human testing: electrode trials on team members first; volunteer sessions only after institutional ethics approval. No medical-device claim is made.

Roadmap: 4-Month Fabrication Plan

M1

Oct 2026 · Fabrication

- Delivery inventory and module revision checks
- Perfboard soldering per Rev A schematic
- ADC bring-up with a signal generator — no body contact yet

M2

Nov 2026 · Fabrication

- PETG print of enclosure + pod on the college printer
- Firmware: timed acquisition + health flags
- Laptop app v0; electrode trials on team members

M3

Dec 2026 · Integration

- Calibration routine and two-factor state machine
- ToF depth gating, IMU motion rejection
- DUALR3 fitted by an electrician and integrated

M4

Jan 2027 · Field demo

- Timed trials: accuracy, false triggers, latency
- Stage 2 → 3 report with measured data
- Demo rehearsal with the faculty guide

Stage 3 Target

- **Commitment:** a working prototype that switches a real fan and light from look + clench, ready for functional testing in Stage 3.
- **Evidence we will bring:** logged accuracy, false-activation and latency numbers from scripted sessions, plus the commits and photos for each milestone.

Dependencies: parts delivery (ordered by 6 Oct 2026), college 3D-printer slot, electrician visit for the DUALR3, and ethics approval before volunteer testing.

Technical Challenges & Risks

Bottleneck: EOG drift vs vergence

- **Constraint:** The near/far vergence change is small next to slow electrode drift. A strong high-pass filter would remove the drift — and the vergence signal with it.
- **If unmanaged:** Wrong depth chosen; selections that wander over a session.

Bottleneck: ESP32 ADC quality

- **Constraint:** The ESP32 ADC is non-linear near its range limits and noisy, and ADC2 is unavailable while Wi-Fi or Bluetooth runs.
- **If unmanaged:** Distorted EOG / EMG traces; poor repeatability between units.

Bottleneck: false jaw-clench triggers

- **Constraint:** Chewing, talking and swallowing all produce masseter EMG that can look like a deliberate clench.
- **If unmanaged:** Unwanted fan or light switching during meals or conversation.

Other risks tracked

- **ToF on dark / glossy targets:** measured surface distance may not match where the eyes converge.
- **No magnetometer:** MPU6500 yaw drifts, so it cannot give a stable room heading.
- **Relay control path:** the DUALR3's local (LAN) vs cloud control options must be confirmed before the app is written.

Why this matters: each of these could make the demo work on the bench but fail with a real user — so each has a planned test on the next slide.

Potential Solutions & Risk Mitigation

Risk	Primary mitigation	Backup / fallback	How we will verify
EOG drift vs vergence	Keep low-frequency path; per-trial baseline removal; discrete near/far targets (as in VergeIO)	Require ToF depth agreement before any candidate	10-min fixed-gaze recordings at both depths
ESP32 ADC quality	ADC1 only, linear input band, 1.6 kHz RC, oversampling, Espressif calibration API	External I2C ADC from the ₹2.9k headroom	Signal-generator sine sweep before body contact
False clench triggers	Clench must be held inside a candidate window; per-user threshold; refractory period	Raise threshold / add dwell; physical disarm	Scripted chew / talk / rest sessions, false triggers counted
ToF on dark / glossy targets	Matte, high-contrast target cards	Use ToF only as a gate, not as gaze depth	Distance tests on matte, dark and glossy cards
IMU yaw drift	Use IMU only to reject head motion	Fixed-head demo posture	Logged yaw during 10-min sessions
Relay control path	Confirm DUALR3 local-LAN control first	Vendor app / Home Assistant bridge	Explicit ON/OFF + state readback test

Verification plan: every risk above has a bench test; results — including failures — will be reported with the Stage 3 submission.

Summary & Stage 3 Readiness

₹13,093
PARTS ORDERED (CAP ₹16,000)

0 / 0
KiCad ERC ERRORS / WARNINGS

FoS ≈ 2.1–2.3
STRAP LOOP, 50 N FEA

4 months
TO A STAGE 3 DEMO

Done (Stage 2)

- Revised architecture inside budget
- All parts ordered from Indian sellers
- 12-part CAD + STEP
- Strap-loop FEA
- Schematic with clean ERC
- Public repository with native files

Next (Oct – Dec 2026)

- Delivery inventory and bring-up
- Enclosure print and assembly
- Firmware + laptop app
- Calibration and two-factor logic
- Relay installation by an electrician

Stage 3 will show

- A real fan and light switched by look + clench
- Measured accuracy, false triggers and latency
- Baseline vs after comparison from the same log format
- Every failure reported, not hidden

Ask: progression to Stage 3 to complete fabrication and measured user testing under faculty supervision.

References & Sources

[1] Zhang X. et al., “VergeIO: Depth-Aware Eye Interaction on Glasses,” arXiv:2507.02187 (2025) — arxiv.org/abs/2507.02187

[2] Upside Down Labs, BioAmp EXG Pill documentation — docs.upsidedownlabs.tech/hardware/bioamp/bioamp-exg-pill

[3] Upside Down Labs, Muscle BioAmp BisCute documentation — docs.upsidedownlabs.tech/hardware/bioamp/muscle-bioamp-biscute

[4] STMicroelectronics, VL53L5CX Time-of-Flight 8×8 multizone sensor — st.com/en/imaging-and-photonics-solutions/vl53l5cx.html

[5] Espressif, ESP-IDF ADC (oneshot) driver and calibration documentation, ESP32 — docs.espressif.com

[6] Microchip, MCP1700 low-quiescent-current LDO datasheet (DS20001826)

[7] SONOFF DUALR3 two-gang smart switch — sonoff.in

[8] International Health Policies, “Spinal cord injuries in India: towards a comprehensive preventive and rehabilitative response”

[9] Office of the Registrar General & Census Commissioner, India — Census 2011 disability data (via punarbhava.in)

[10] SCIRE Project, Non-intervention attendant care studies (Robinson-Whelan & Rintala 2003) — scireproject.com

[11] Amrita Vishwa Vidyapeetham, news release on Netravaad eye-gesture communication device — amrita.edu

Data sources: statistics on slides 2–4 and the Netravaad price come from [8]–[11]; component specifications from [2]–[7].

Image Sources

Slide	Figure	Source / tool
4	Before-state user flow	Project diagram (drawn for this submission)
5	After-state user flow	Project diagram (drawn for this submission)
6	System block diagram	Project diagram (drawn for this submission)
8	Exploded and assembled enclosure views	FreeCAD 1.1 — cad/VergeLink_Enclosure.FCStd
9	Strap-loop von Mises stress plot	FreeCAD FEM + CalculiX — cad/VergeLink_StrapLoop_FEA.FCStd
10	Mainboard schematic	KiCad 10 — kicad/VergeLink_Mainboard.kicad_sch
12	FreeCAD session screenshot	Screen capture of the project model

Third-party images: none. All files are in the project repository: github.com/amalnathkp13-boop/vergelink